Predikcija rizika od kardiobaskularnih bolesti

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Projekat iz predmeta "Veštačka inteligenicja u medicini"

Dataset: https://www.kaggle.com/datasets/bertnardomariouskono/cardiovascular-disease-risk-prediction-dataset

Uvod

Nad pomenutim datasetom iskorišćene su tehnike *ML*-a u cilju detektovanja i određivanje verovatnoće rizika oboljenja od kardiovaskularnih bolsti. Kardiovaskularne bolesti su oboljenja koja zahvataju srce i krvne sudove (arterije i vene). To su najčešći uzrok smrti u svetu, ali se mnoge mogu sprečiti zdravim načinom života i redovnim kontrolama. Konkretna tehnika koja je korišćena je binarna klasifikacija. U projektu je obrađeno više algoritama binarne klasifikacije koji će biti upoređeni. Rad se sastoji od preprocesiranja dataseta, primena algoritama i vizualizacija rezultata uz pomoć nekih validacionih metoda. Na kraju, rezultati će biti upoređeni sa *finetune*-ovanim algoritmima koji su se najbolje pokazali.

Dataset

- Patient_ID - Id pacijenta. Polje će biti ignorisano
- Age - Godina starosti
- Gender - Pol
- Height_cm - Visina
- Weight_kg - Težina
- BMI
- Systolic_BP
- Diastolic_BP
- Cholesterol_Total
- Cholesterol_LDL
- Cholesterol_HDL
- Fasting_Blood_Sugar
- Smoking_Status
- Alcohol_Consumption
- Physical_Activity_Level
- Family_History
- Stress_Level
- Sleep_Hours
- Heart_Disease_Risk

Kod

Library import

In [ ]:
```
from sklearn.model_selection import cross_validate, train_test_split, Str
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClas
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import confusion_matrix
from sklearn.metrics import recall_score
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import confusion_matrix, roc_curve, auc
from sklearn.preprocessing import label_binarize
from sklearn.metrics import make_scorer, recall_score
from sklearn.metrics import (
    recall_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
    roc_curve,
    auc
)
import matplotlib.pyplot as plt
from sklearn.ensemble import IsolationForest
from sklearn.pipeline import Pipeline
import seaborn as sn
import pandas as pd
import numpy as np

sn.set_theme(style="whitegrid", context="talk")
plt.rcParams["figure.figsize"] = (12, 6)
```

Helper functions

In [2]:
```
class ModelResult:
    def __init__(self, name, cv_metrics, y_pred_test=None, y_proba_test=N
        self.name = name
        self.cv_metrics = cv_metrics
        self.y_pred_test = y_pred_test
        self.y_proba_test = y_proba_test

results_list = []  # Rezultati svakog algoritma
tuned_parameters_results_list = [] # Rezultati algoritama posle fine-tuni

def evaluate_cv(pipeline, X_train, y_train, cv):
    scores = cross_validate(
        pipeline,
        X_train,
        y_train,
        cv=cv,
        scoring={
            'accuracy': 'accuracy',
            'macro_f1': 'f1_macro',
```

```
            'macro_recall': 'recall_macro'
        }
    )

    return {
        "accuracy": scores['test_accuracy'].mean(),
        "macro_f1": scores['test_macro_f1'].mean(),
        "macro_recall": scores['test_macro_recall'].mean()
    }

def train_and_predict(pipeline, X_train, y_train, X_test):
    pipeline.fit(X_train, y_train)
    y_pred = pipeline.predict(X_test)
    y_proba = pipeline.predict_proba(X_test)
    return y_pred, y_proba

def plot_confusion_matrix(y_true, y_pred, labels, title):
    cm = confusion_matrix(y_true, y_pred)
    disp = ConfusionMatrixDisplay(cm, display_labels=labels)

    plt.figure(figsize=(8, 6))
    disp.plot(cmap="Blues", values_format="d")
    plt.title(title)
    plt.show()

def plot_roc(y_true, y_proba, classes, title):
    plt.figure()

    if len(classes) == 2:
        proba = y_proba[:, 1] if y_proba.ndim > 1 else y_proba
        fpr, tpr, _ = roc_curve(y_true, proba)
        roc_auc = auc(fpr, tpr)
        plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
    else:
        y_bin = label_binarize(y_true, classes=classes)
        roc_auc = {}
        for i, cls in enumerate(classes):
            fpr, tpr, _ = roc_curve(y_bin[:, i], y_proba[:, i])
            roc_auc[cls] = auc(fpr, tpr)
            plt.plot(fpr, tpr, label=f"Class {cls} AUC = {roc_auc[cls]:.3

    plt.plot([0, 1], [0, 1], '--', color='gray')
    plt.xlabel("False Positive Rate")
    plt.ylabel("True Positive Rate")
    plt.title(title)
    plt.legend()
    plt.tight_layout()
    plt.show()

    return roc_auc

def plot_cv_metrics(results_list, metrics=["accuracy","macro_f1","macro_r

    data = []
    for res in results_list:
        row = {"Algorithm": res.name}
        for m in metrics:
            row[m] = res.cv_metrics[m]
        data.append(row)
```

```
    df = pd.DataFrame(data)
    df_melted = df.melt(id_vars="Algorithm", var_name="Metric", value_nam

    plt.figure(figsize=(10,6))
    ax = sn.barplot(data=df_melted, x="Algorithm", y="Score", hue="Metric

    plt.ylim(0, 1.05)
    plt.title("CV Metrics Comparison")
    plt.xticks(rotation=45)
    plt.ylabel("Score")
    plt.legend(loc="lower right")

    for p in ax.patches:
        height = p.get_height()
        x = p.get_x() + p.get_width() / 2

        if height > 0.15:
            va = "top"
            y = height - 0.02
            color = "white"
            weight = "bold"
        else:
            va = "bottom"
            y = height + 0.02
            color = "black"
            weight = "normal"

        ax.text(x, y, f"{height:.3f}", ha="center", va=va, color=color, f

    plt.tight_layout()
    plt.show()

def plot_feature_importance(model_pipeline, feature_names, top_n=None, ti
    if hasattr(model_pipeline, "named_steps"):
        steps = list(model_pipeline.named_steps.keys())
        model = model_pipeline.named_steps[steps[-1]]

        for step_name in steps[:-1]:
            step = model_pipeline.named_steps[step_name]
            if hasattr(step, 'get_feature_names_out'):
                try:
                    feature_names = list(step.get_feature_names_out())
                except Exception:
                    pass
    else:
        model = model_pipeline

    if hasattr(model, "feature_importances_"):
        importances = model.feature_importances_
    elif hasattr(model, "coef_"):
        importances = np.mean(np.abs(model.coef_), axis=0) if model.coef_

    if len(feature_names) != len(importances):
        feature_names = [f"feature_{i}" for i in range(len(importances))]

    importance_df = pd.DataFrame({
        "feature": feature_names,
        "importance": importances
    }).sort_values(by="importance", ascending=False)
```

```
    if top_n:
        importance_df = importance_df.head(top_n)

    plt.figure()
    plt.barh(importance_df["feature"], importance_df["importance"])
    plt.gca().invert_yaxis()
    plt.title(title if title else "Feature Importance")
    plt.xlabel("Importance")
    plt.tight_layout()
    plt.show()

    return importance_df

def plot_correlation_matrix(X, numerical_features, title="Feature Correla
    corr = X[numerical_features].corr()

    mask = np.triu(np.ones_like(corr, dtype=bool), k=1)

    plt.figure(figsize=(14, 10))
    sn.heatmap(
        corr,
        mask=mask,
        annot=True,
        fmt=".1f",
        cmap="coolwarm",
        center=0,
        square=True,
        linewidths=0.5,
        annot_kws={"size": 8}
    )
    plt.title(title, fontsize=14)
    plt.xticks(rotation=45, ha='right', fontsize=9)
    plt.yticks(fontsize=9)
    plt.tight_layout()
    plt.show()
```

Data import

In [3]:
```
dataframe = pd.read_csv('healthcare_synthetic_data.csv')

dataframe.head(15)
```

Out[3]:

	Patient_ID	Age	Gender	Height_cm	Weight_kg	BMI	Systolic_BP	Diastolic_BP
0	PID-00001	60	0	146.9	51.3	23.8	140	89
1	PID-00002	53	0	161.8	76.6	29.3	128	81
2	PID-00003	62	1	174.7	92.4	30.3	141	100
3	PID-00004	73	1	173.3	68.9	22.9	136	96
4	PID-00005	52	1	178.6	79.8	25.0	122	80
5	PID-00006	52	0	159.6	60.3	23.7	134	92
6	PID-00007	73	0	158.2	62.7	25.0	149	98
7	PID-00008	64	0	144.5	72.0	34.5	129	90
8	PID-00009	49	0	168.7	80.2	28.2	139	91
9	PID-00010	61	1	176.8	113.6	36.3	131	91
10	PID-00011	49	0	157.4	83.9	33.9	150	100
11	PID-00012	49	0	162.2	84.5	32.1	141	91
12	PID-00013	57	0	158.7	84.9	33.7	126	90
13	PID-00014	32	0	161.2	83.8	32.3	137	92
14	PID-00015	34	1	179.6	112.9	35.0	124	89

In [4]:

```
print(dataframe.count)
```

```
<bound method DataFrame.count of       Patient_ID  Age  Gender  Height_cm
Weight_kg   BMI  Systolic_BP  \
0      PID-00001   60       0      146.9       51.3  23.8          140
1      PID-00002   53       0      161.8       76.6  29.3          128
2      PID-00003   62       1      174.7       92.4  30.3          141
3      PID-00004   73       1      173.3       68.9  22.9          136
4      PID-00005   52       1      178.6       79.8  25.0          122
...          ...  ...     ...        ...        ...   ...          ...
14995  PID-14996   44       1      166.9       81.8  29.4          135
14996  PID-14997   69       0      154.0       61.7  26.0          124
14997  PID-14998   70       1      178.5       88.6  27.8          129
14998  PID-14999   44       0      157.3       60.5  24.5          122
14999  PID-15000   53       1      170.1       72.7  25.1          150

       Diastolic_BP  Cholesterol_Total  Cholesterol_LDL  Cholesterol_HDL
\
0                89                217              151               52
1                81                203              119               38
2               100                173              124               45
3                96                193              117               45
4                80                236              153               41
...             ...                ...              ...              ...
14995            85                203              123               33
14996            79                219              147               57
14997            93                231              131               45
14998            84                185              133               44
14999           101                215              130               44

       Fasting_Blood_Sugar  Smoking_Status  Alcohol_Consumption  \
0                       83               0                    1
1                      116               0                    0
2                       90               0                    0
3                       81               0                    0
4                       79               0                    1
...                    ...             ...                  ...
14995                   75               0                    1
14996                  114               0                    1
14997                  105               0                    1
14998                  105               1                    2
14999                   72               1                    1

       Physical_Activity_Level  Family_History  Stress_Level  Sleep_Hours
\
0                            3               0             1            8
1                            1               0             7            9
2                            0               0             1            7
3                            1               0             2            7
4                            2               0             2            6
...                        ...             ...           ...          ...
14995                        2               0             9            6
14996                        1               0             6            8
14997                        0               1             9            6
14998                        2               0             4            8
14999                        1               0             6            6

       Heart_Disease_Risk
0                       0
1                       0
2                       1
3                       1
```

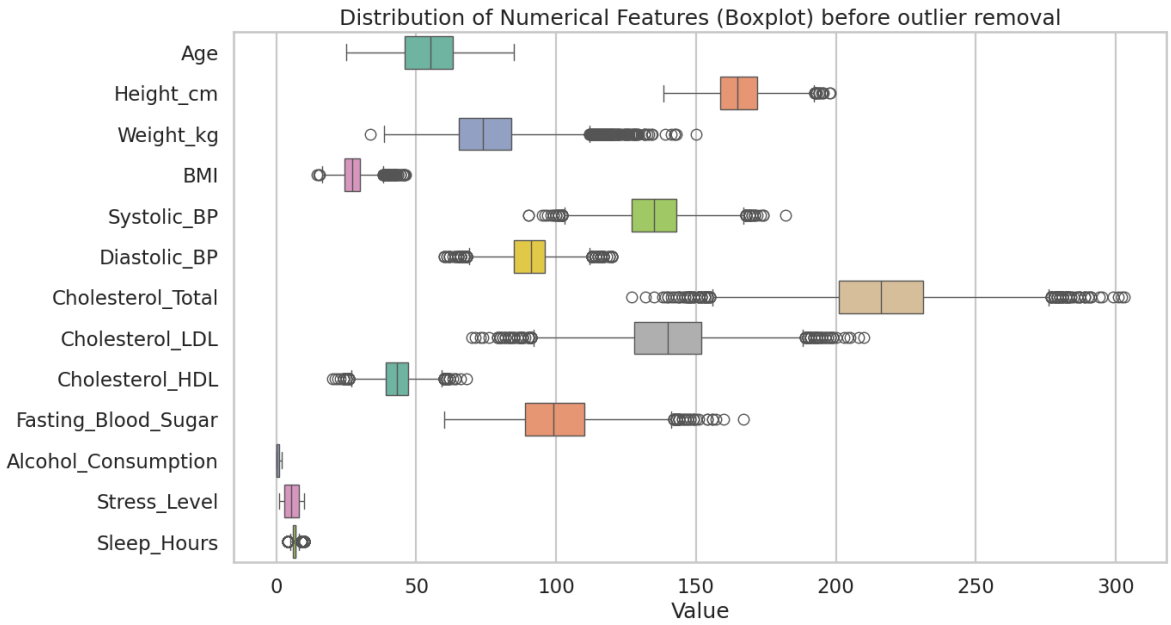
```
4                        0
...                    ...
14995                    1
14996                    1
14997                    1
14998                    1
14999                    1

[15000 rows x 19 columns]>
```

Data preprocessing

In [5]:
```
# Brisanje podataka sa null vrednostima, augmentacija u ovom datasetu nij
dataframe.dropna(inplace=True)

# ID pacijenta nam nije potreban
dataframe = dataframe.drop("Patient_ID", axis=1)

categorical = ['Gender', "Smoking_Status", "Physical_Activity_Level", "Fa
numerical = ['Age', 'Height_cm', 'Weight_kg', 'BMI', 'Systolic_BP', 'Dias
feature_names = categorical + numerical

X = dataframe[numerical]
```

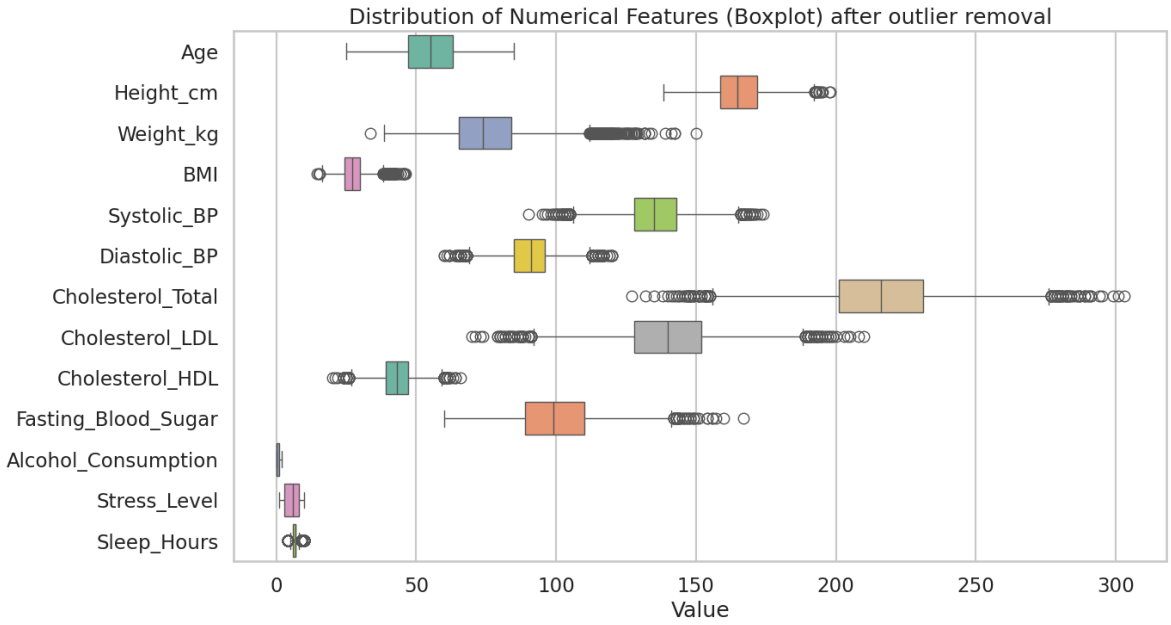
Outlier detection

In [6]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

iso = IsolationForest(
    n_estimators=300,
    contamination=0.005,
    random_state=42
)

outliers = iso.fit_predict(X_scaled)

plt.figure(figsize=(14, 8))
sn.boxplot(data=dataframe[numerical], orient="h", palette="Set2")
plt.title("Distribution of Numerical Features (Boxplot) before outlier re
plt.xlabel("Value")
plt.show()
```



Outlier removal

In [7]:
```
dataframe = dataframe[outliers == 1]

plt.figure(figsize=(14, 8))
sn.boxplot(data=dataframe[numerical], orient="h", palette="Set2")
plt.title("Distribution of Numerical Features (Boxplot) after outlier rem
plt.xlabel("Value")
plt.show()

print(f"Removed {np.sum(outliers==-1)} outliers ({np.mean(outliers==-1)*1
```

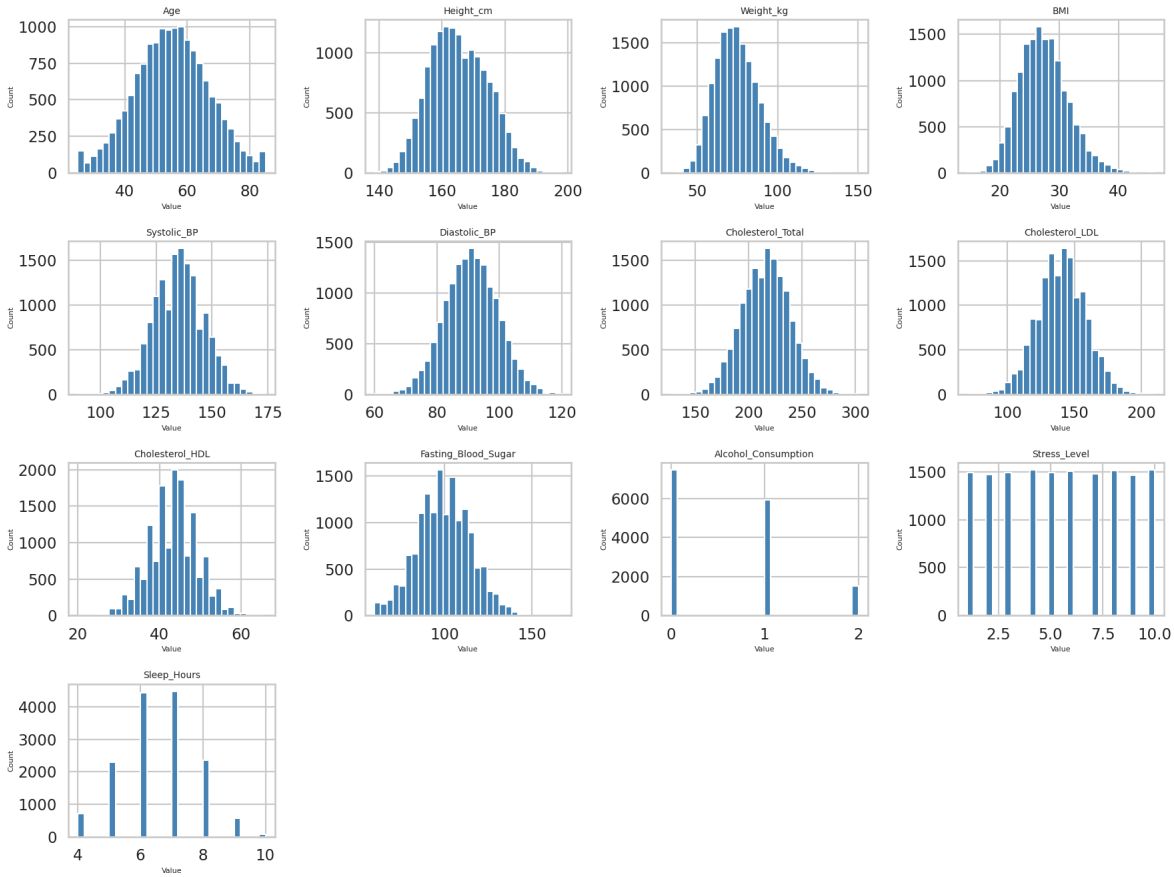



```
Removed 75 outliers (0.50%)
```

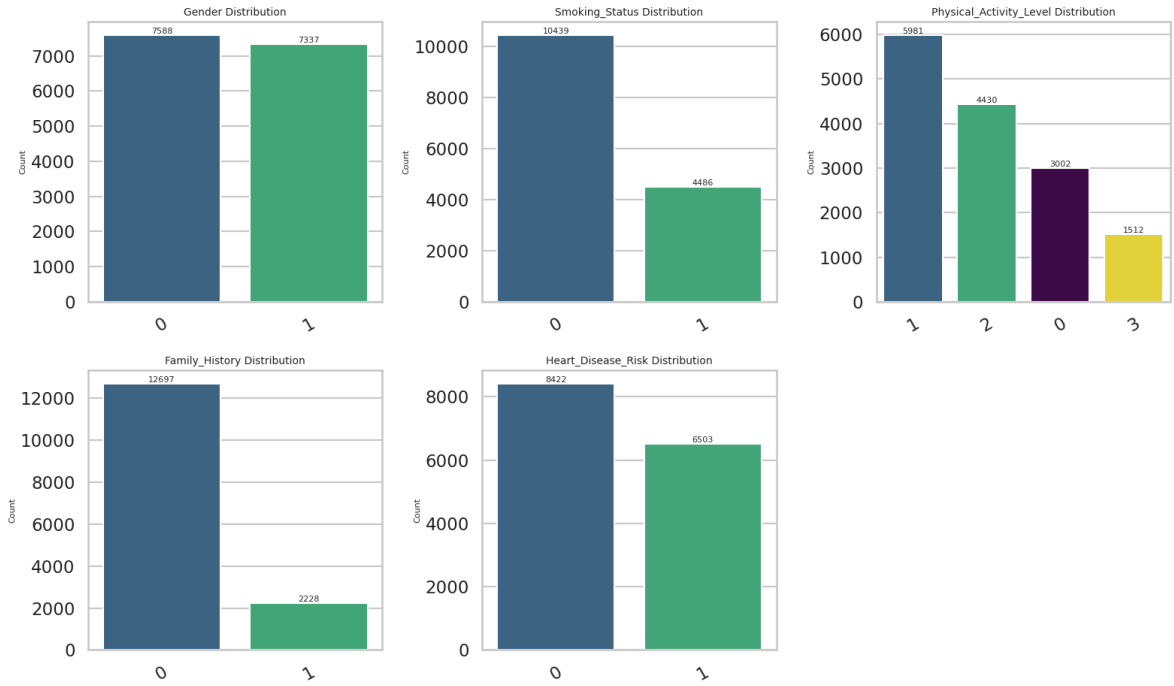
In [8]:
```
fig, axes = plt.subplots(nrows=4, ncols=4, figsize=(18, 14))
axes = axes.flatten()
for i, col in enumerate(numerical):
    axes[i].hist(dataframe[col], bins=30, color='steelblue', edgecolor='w
    axes[i].set_title(col, fontsize=10)
    axes[i].set_xlabel("Value", fontsize=8)
    axes[i].set_ylabel("Count", fontsize=8)
```

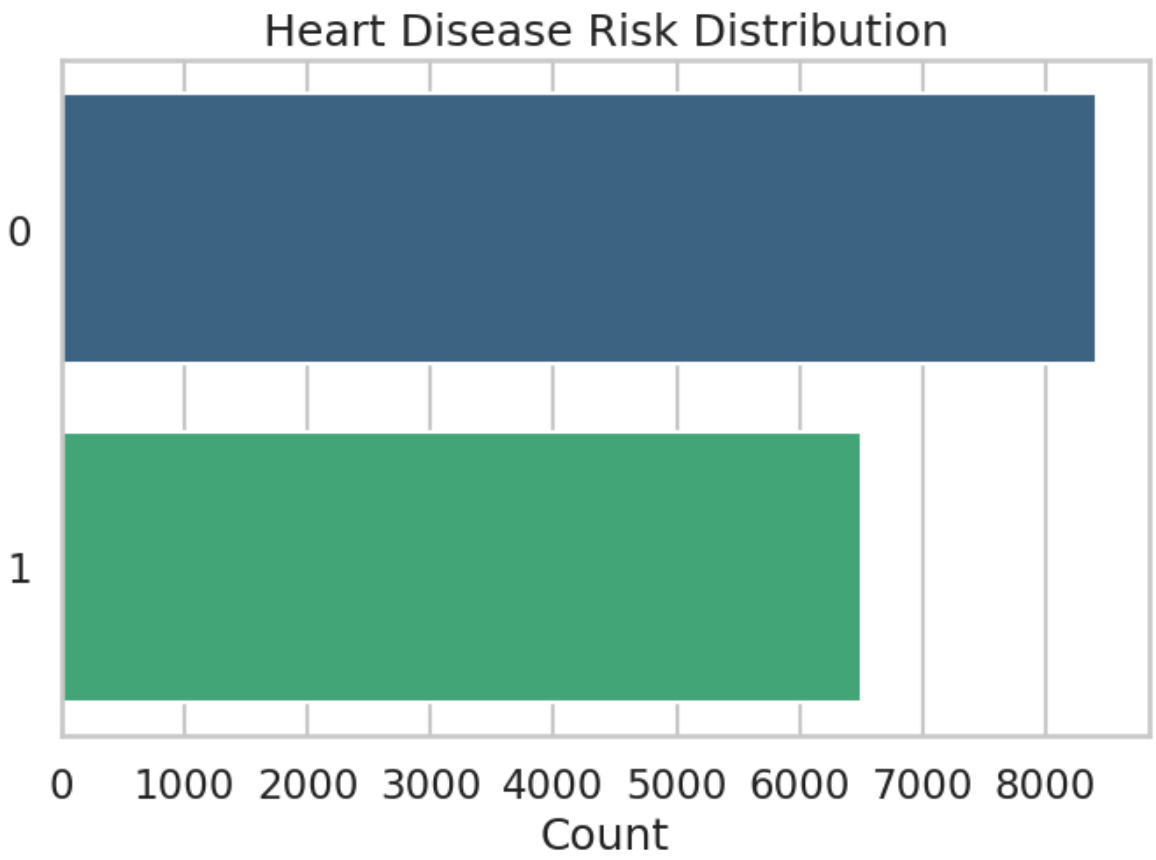
```
for j in range(len(numerical), len(axes)):
    axes[j].set_visible(False)

plt.suptitle("Numerical Feature Distributions (Histograms)", fontsize=14,
plt.tight_layout()
plt.show()

fig, axes = plt.subplots(nrows=2, ncols=3, figsize=(16, 10))
axes = axes.flatten()
for i, col in enumerate(categorical):
    order = dataframe[col].value_counts().index
    sn.countplot(data=dataframe, x=col, hue=col, order=order, palette='vi
    axes[i].set_title(f"{col} Distribution", fontsize=10)
    axes[i].set_xlabel("")
    axes[i].set_ylabel("Count", fontsize=8)
    axes[i].tick_params(axis='x', rotation=30)


    for p in axes[i].patches:
        axes[i].annotate(f'{int(p.get_height())}',
                         (p.get_x() + p.get_width() / 2, p.get_height()),
                         ha='center', va='bottom', fontsize=8)


for j in range(len(categorical), len(axes)):
    axes[j].set_visible(False)

plt.suptitle("Categorical Feature Distributions", fontsize=14)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sn.countplot(
    y='Heart_Disease_Risk',
    hue='Heart_Disease_Risk',
    data=dataframe,
    order=dataframe['Heart_Disease_Risk'].value_counts().index,
    palette='viridis',
    legend=False
)
plt.title("Heart Disease Risk Distribution")
plt.xlabel("Count")
plt.ylabel("")
plt.show()
```

Numerical Feature Distributions (Histograms)



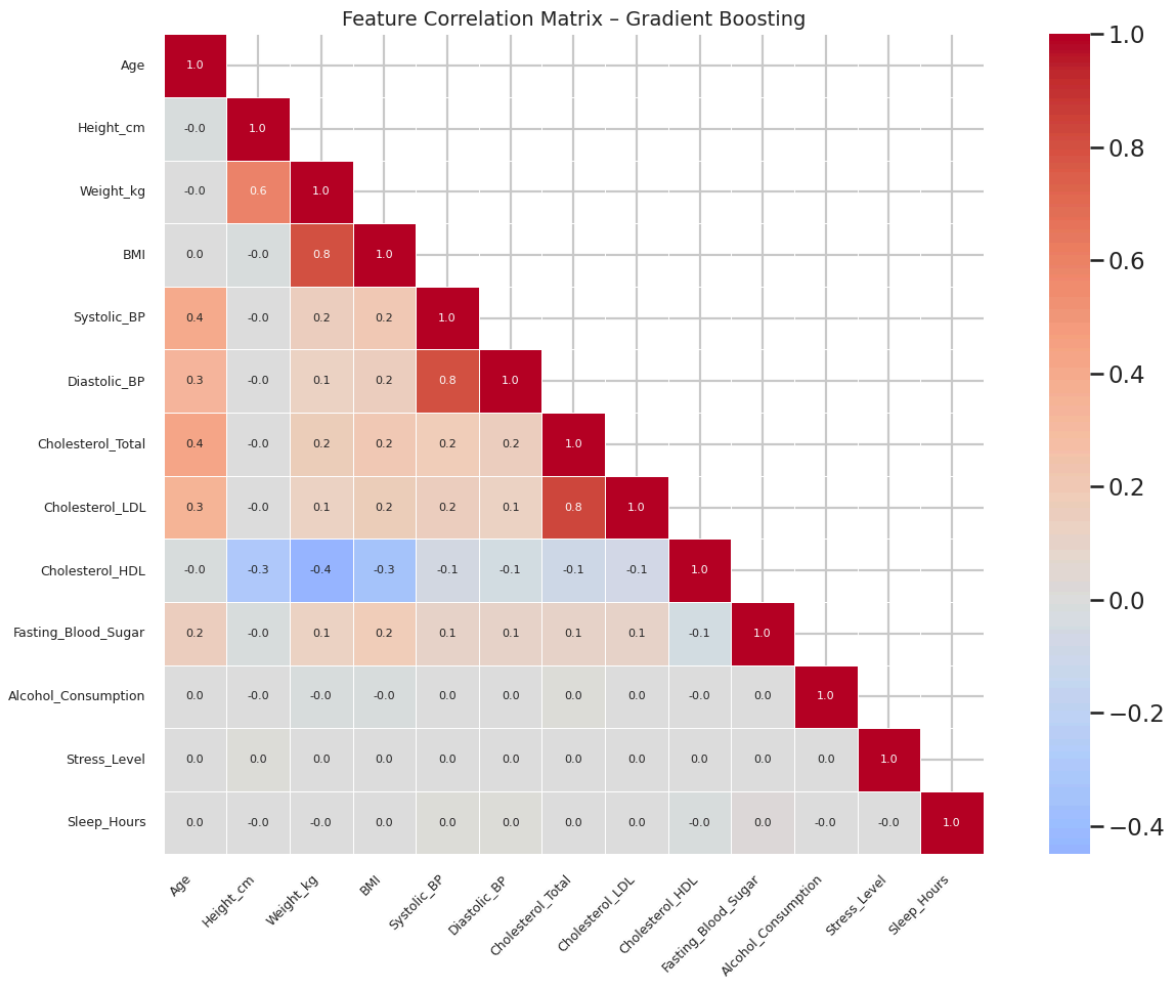
Categorical Feature Distributions





Train-Test split, izdvojićemo test podskup za finalno testiranje modela na kraju

In [9]:
```
X = dataframe.drop('Heart_Disease_Risk', axis=1)
y = dataframe['Heart_Disease_Risk']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    stratify=y,
    random_state=42
)

plot_correlation_matrix(dataframe, numerical, "Feature Correlation Matrix
```



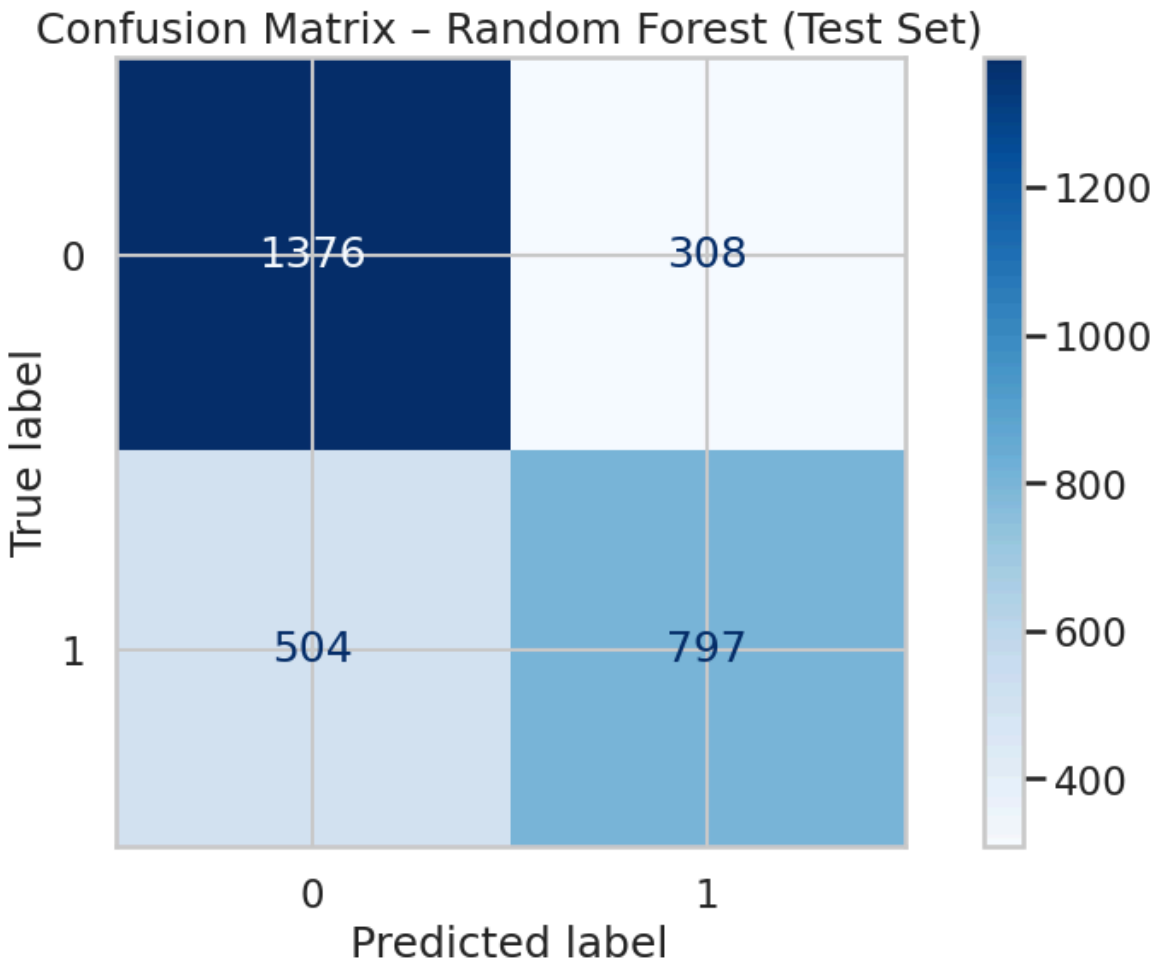
Testiranje različitih algoritama multiklasne klasifikacije

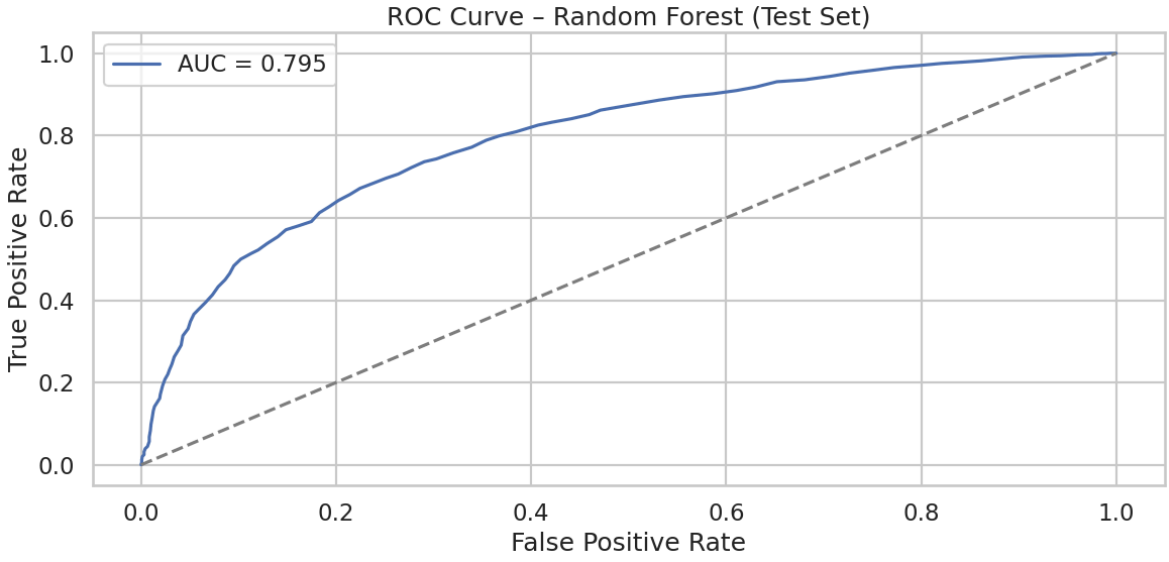
RandomForestClassifier

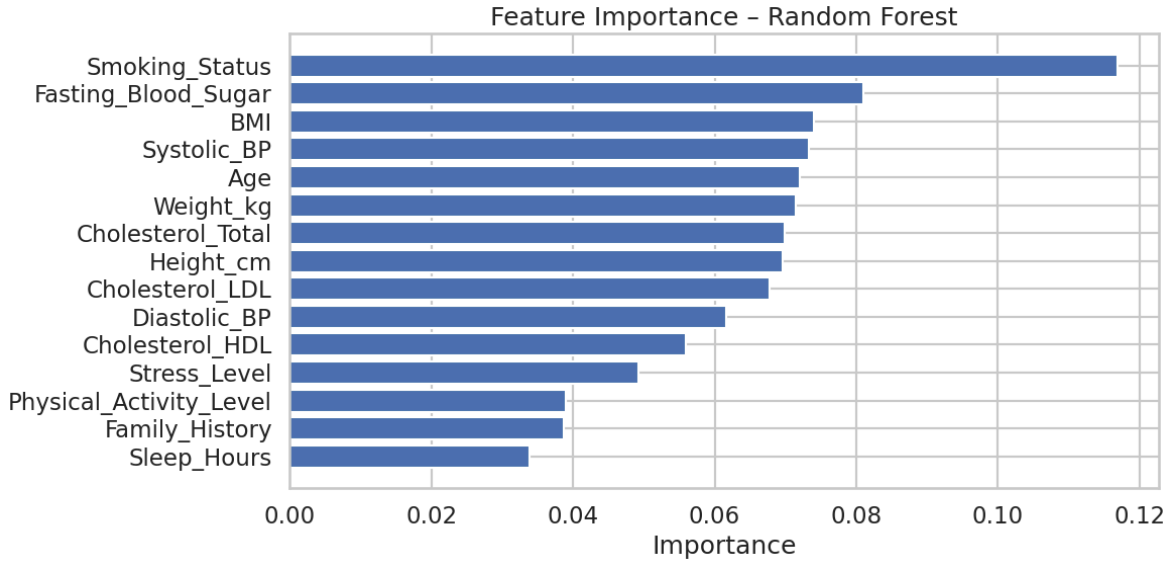
In [10]:
```
pipeline_rf = Pipeline([
    ('scaler', StandardScaler()),
    ('model', RandomForestClassifier())
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_results = evaluate_cv(pipeline_rf, X_train, y_train, cv)

print("Random Forest (CV mean):")
for k, v in cv_results.items():
    print(f"{k.capitalize():15}: {v:.3f}")

y_pred_test, y_proba_test = train_and_predict(
    pipeline_rf, X_train, y_train, X_test
)

classes = np.unique(y_train)

plot_confusion_matrix(
    y_test, y_pred_test, classes,
    "Confusion Matrix – Random Forest (Test Set)"
)
```

```
plot_roc(
    y_test, y_proba_test, classes,
    "ROC Curve — Random Forest (Test Set)"
)

plot_feature_importance(pipeline_rf, feature_names, top_n=15, title="Feat

rf_result = ModelResult(
    name="Random Forest",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

results_list.append(rf_result)
```

```
Random Forest (CV mean):
Accuracy        : 0.723
Macro_f1        : 0.713
Macro_recall    : 0.711
<Figure size 800x600 with 0 Axes>
```





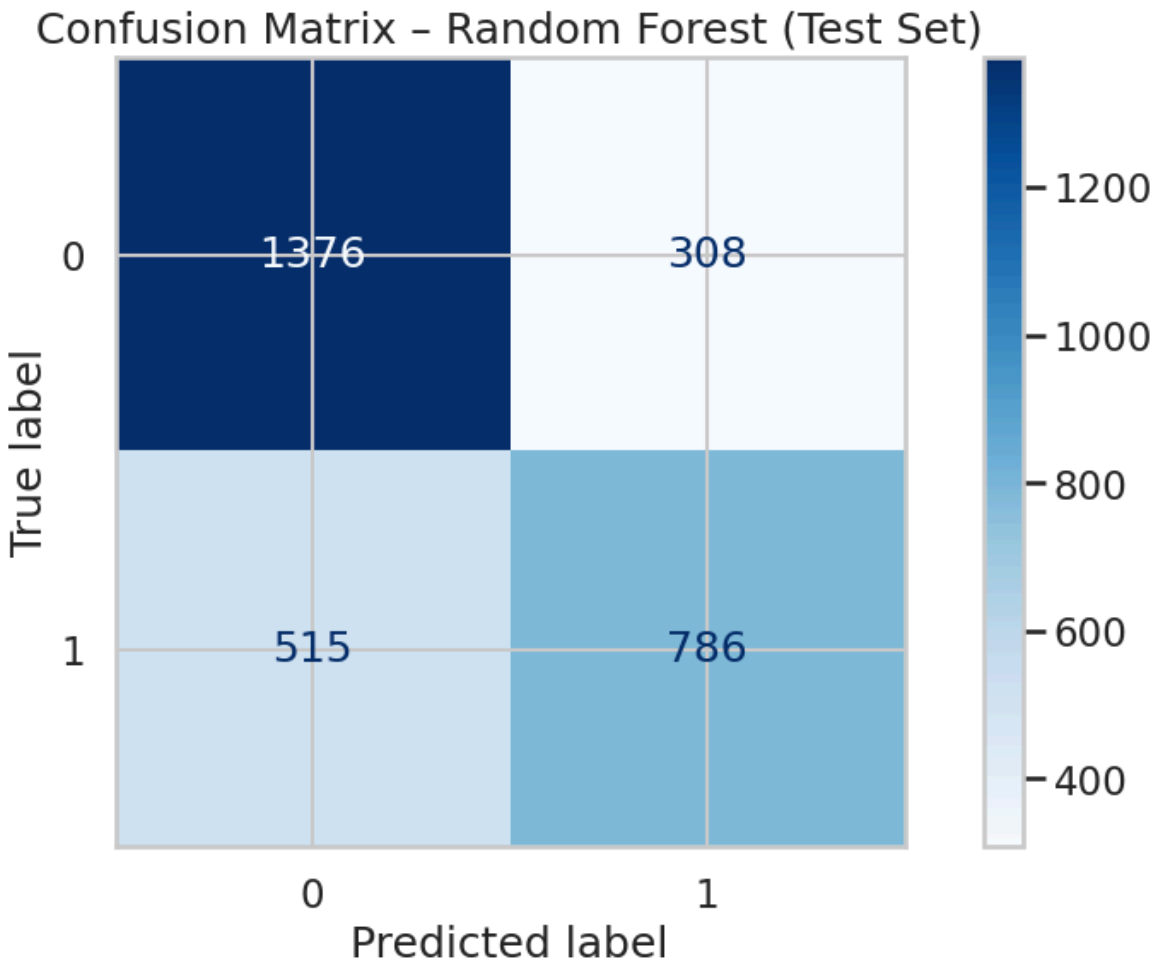


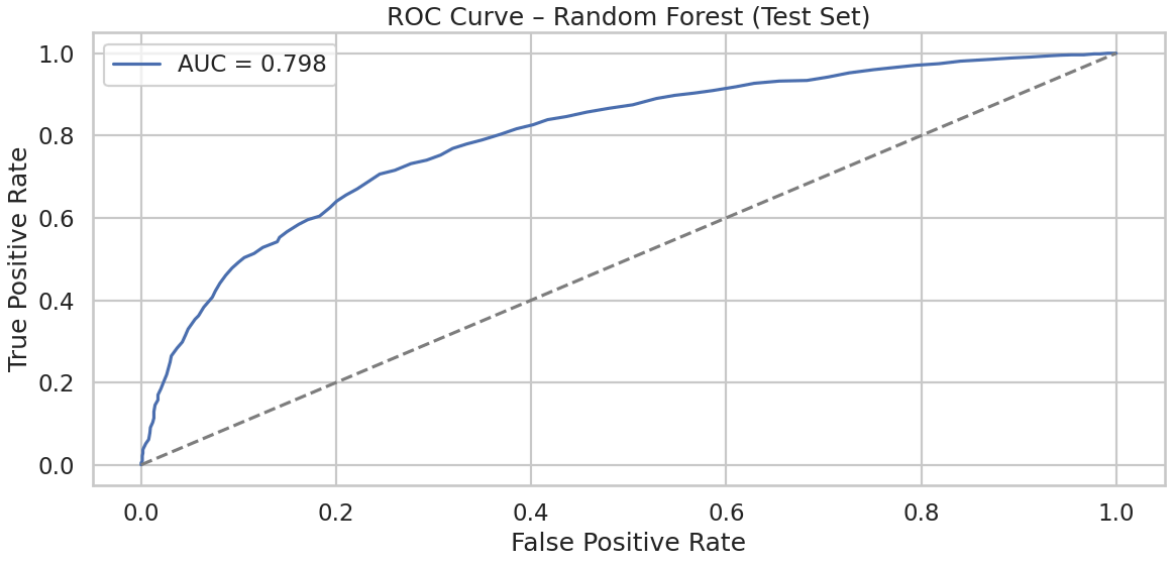
In [11]:

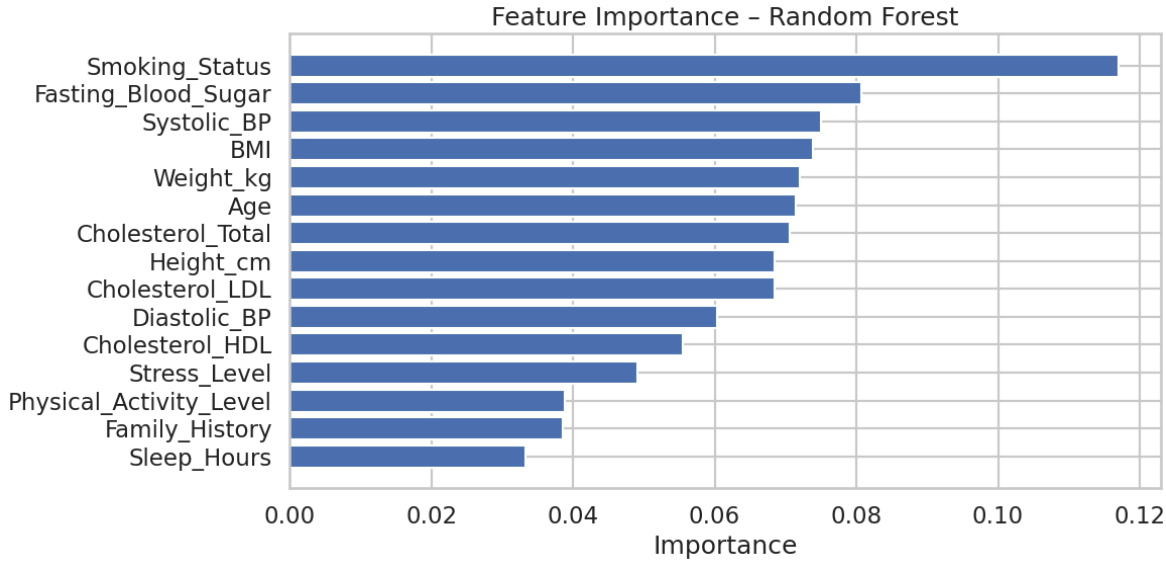
```
pipeline_rf = Pipeline([
    ('scaler', StandardScaler()),
    ('model', RandomForestClassifier())
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_results = evaluate_cv(pipeline_rf, X_train, y_train, cv)

print("Random Forest (CV mean):")
for k, v in cv_results.items():
    print(f"{k.capitalize():15}: {v:.3f}")

y_pred_test, y_proba_test = train_and_predict(
    pipeline_rf, X_train, y_train, X_test
)

classes = np.unique(y_train)

plot_confusion_matrix(
    y_test, y_pred_test, classes,
    "Confusion Matrix — Random Forest (Test Set)"
)

plot_roc(
```

```
    y_test, y_proba_test, classes,
    "ROC Curve – Random Forest (Test Set)"
)

plot_feature_importance(pipeline_rf, feature_names, top_n=15, title="Feat

rf_result = ModelResult(
    name="Random Forest",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

results_list.append(rf_result)
```

```
Random Forest (CV mean):
Accuracy       : 0.724
Macro_f1       : 0.714
Macro_recall   : 0.712
<Figure size 800x600 with 0 Axes>
```





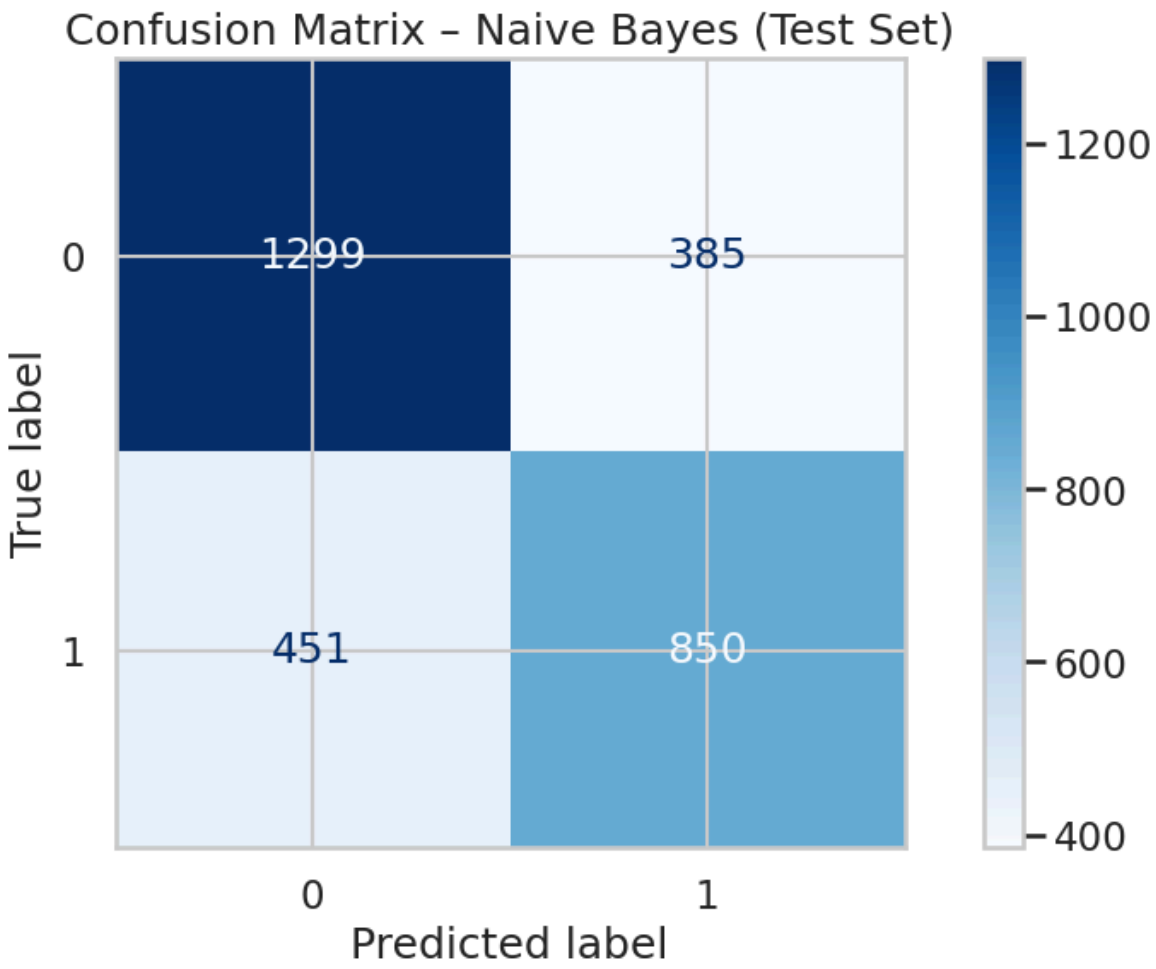


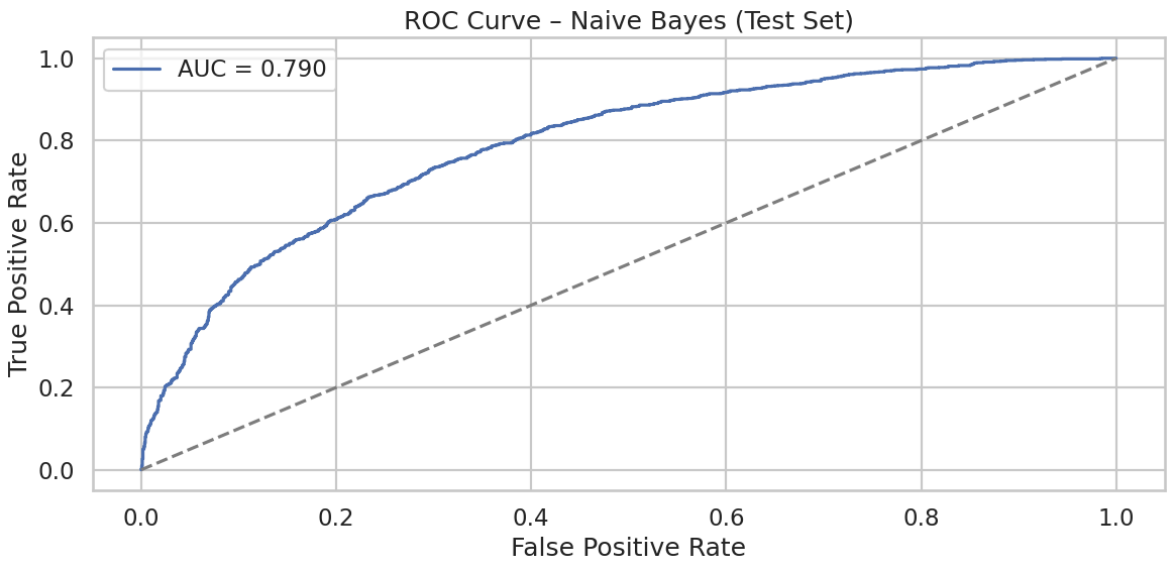
Naive Bayes

In [12]:
```
pipeline_rf = Pipeline([
    ('scaler', StandardScaler()),
    ('model', GaussianNB())
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_results = evaluate_cv(pipeline_rf, X_train, y_train, cv)

print("Naive Bayes (CV mean):")
for k, v in cv_results.items():
    print(f"{k.capitalize():15}: {v:.3f}")

y_pred_test, y_proba_test = train_and_predict(
    pipeline_rf, X_train, y_train, X_test
)

classes = np.unique(y_train)

plot_confusion_matrix(
    y_test, y_pred_test, classes,
    "Confusion Matrix — Naive Bayes (Test Set)"
```

```
)

plot_roc(
    y_test, y_proba_test, classes,
    "ROC Curve – Naive Bayes (Test Set)"
)


rf_result = ModelResult(
    name="Naive Bayes",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

results_list.append(rf_result)
```

```
Naive Bayes (CV mean):
Accuracy       : 0.714
Macro_f1       : 0.709
Macro_recall   : 0.708
<Figure size 800x600 with 0 Axes>
```

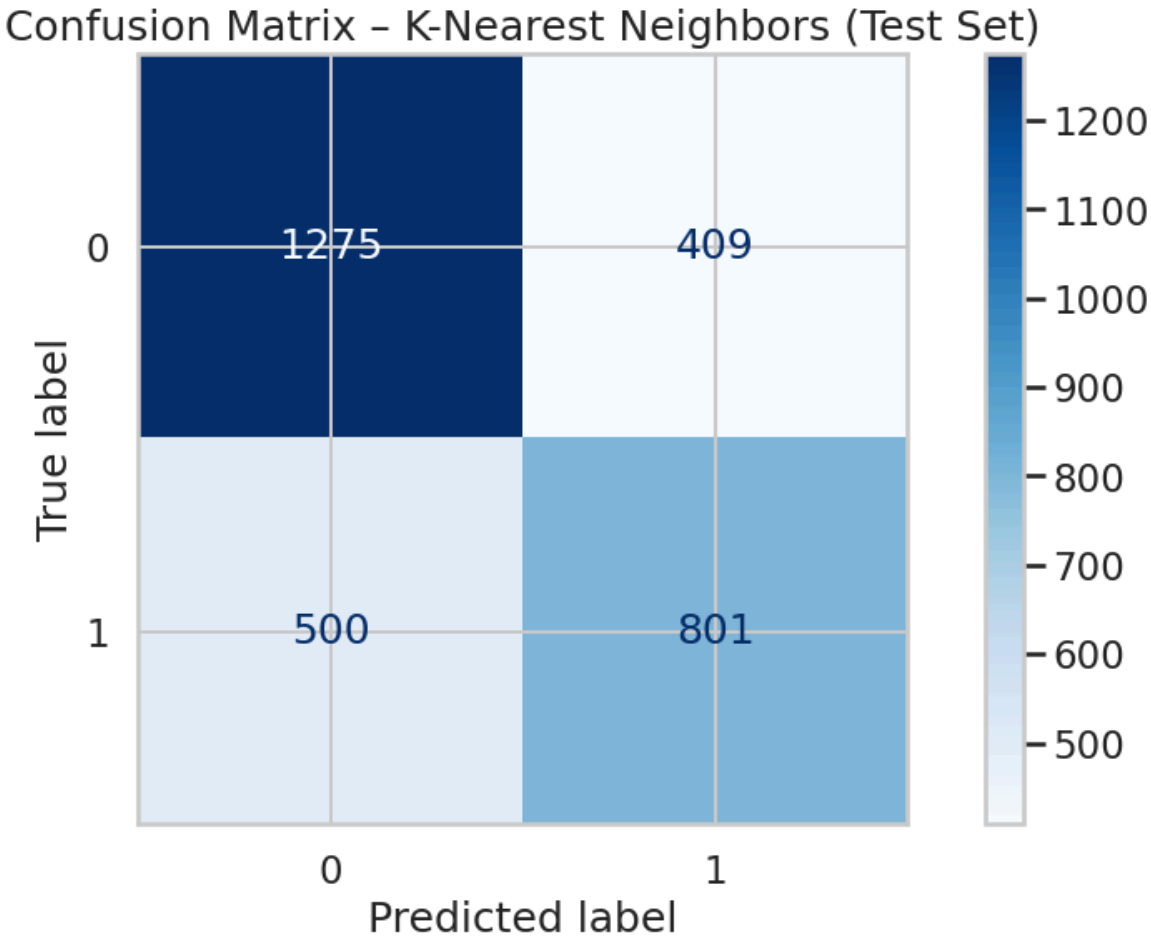


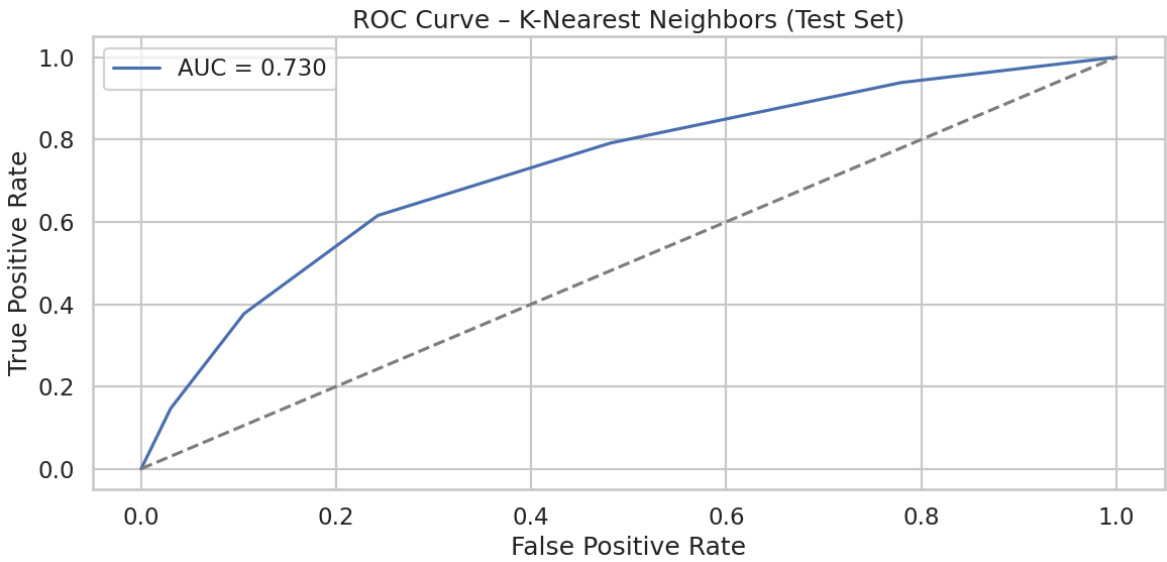


K-Nearest Neighbors

In [13]:
```
pipeline_rf = Pipeline([
    ('scaler', StandardScaler()),
    ('model', KNeighborsClassifier())
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_results = evaluate_cv(pipeline_rf, X_train, y_train, cv)

print("K-Nearest Neighbors (CV mean):")
for k, v in cv_results.items():
    print(f"{k.capitalize():15}: {v:.3f}")

y_pred_test, y_proba_test = train_and_predict(
    pipeline_rf, X_train, y_train, X_test
)

classes = np.unique(y_train)

plot_confusion_matrix(
    y_test, y_pred_test, classes,
    "Confusion Matrix — K-Nearest Neighbors (Test Set)"
)

plot_roc(
    y_test, y_proba_test, classes,
    "ROC Curve — K-Nearest Neighbors (Test Set)"
)


rf_result = ModelResult(
    name="K-Nearest Neighbors",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

results_list.append(rf_result)
```

```
K-Nearest Neighbors (CV mean):
Accuracy       : 0.682
Macro_f1       : 0.674
Macro_recall   : 0.673
<Figure size 800x600 with 0 Axes>
```



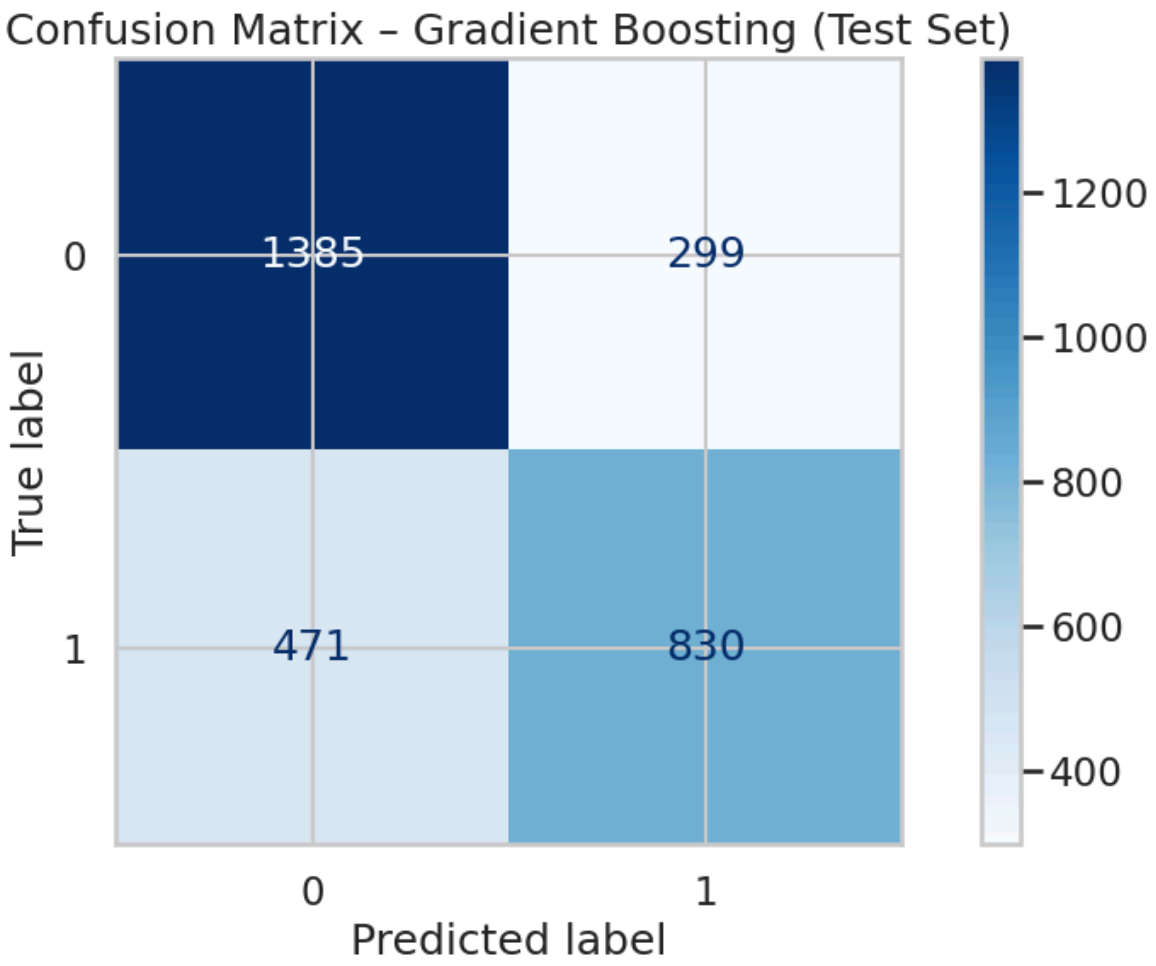


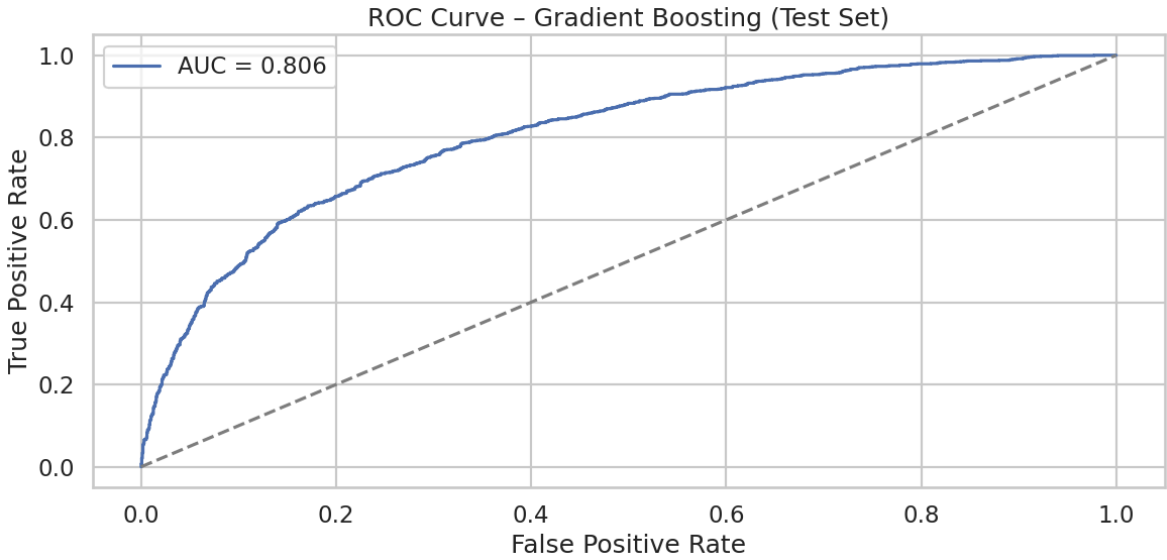
Gradient Boosting

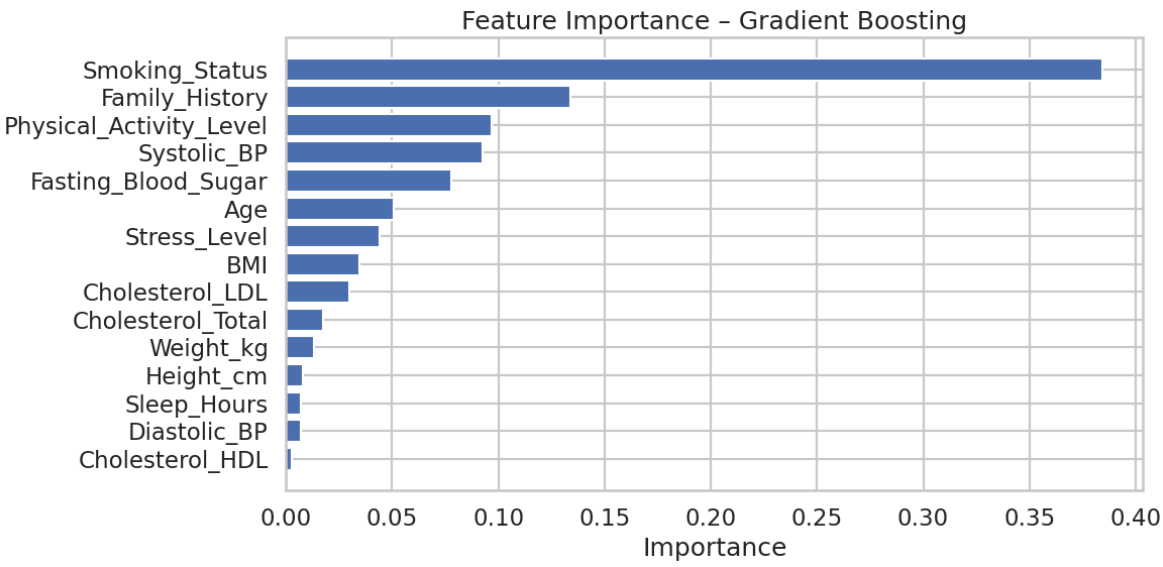
In [14]:
```
pipeline_rf = Pipeline([
    ('scaler', StandardScaler()),
    ('model', GradientBoostingClassifier())
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

```python
cv_results = evaluate_cv(pipeline_rf, X_train, y_train, cv)

print("Gradient Boosting (CV mean):")
for k, v in cv_results.items():
    print(f"{k.capitalize():15}: {v:.3f}")

y_pred_test, y_proba_test = train_and_predict(
    pipeline_rf, X_train, y_train, X_test
)

classes = np.unique(y_train)

plot_confusion_matrix(
    y_test, y_pred_test, classes,
    "Confusion Matrix — Gradient Boosting (Test Set)"
)

plot_roc(
    y_test, y_proba_test, classes,
    "ROC Curve — Gradient Boosting (Test Set)"
)

plot_feature_importance(pipeline_rf, feature_names, top_n=15, title="Feat

rf_result = ModelResult(
    name="Gradient Boosting",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

results_list.append(rf_result)
```

```
Gradient Boosting (CV mean):
Accuracy       : 0.734
Macro_f1       : 0.725
Macro_recall   : 0.723
<Figure size 800x600 with 0 Axes>
```


Confusion Matrix – Gradient Boosting (Test Set)
True label
Predicted label
0
1
0
1
1385
299
471
830
1200
1000
800
600
400

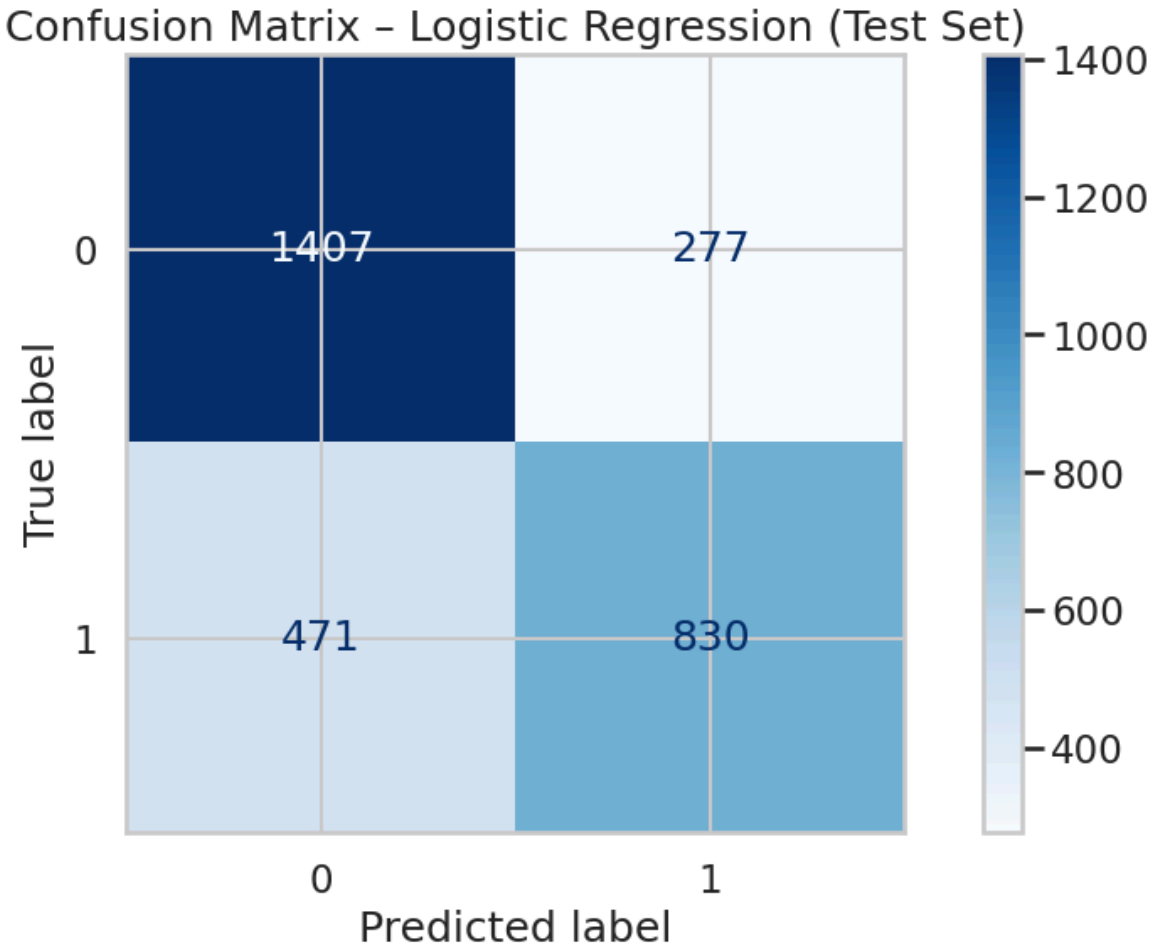

ROC Curve – Gradient Boosting (Test Set)
AUC = 0.806
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

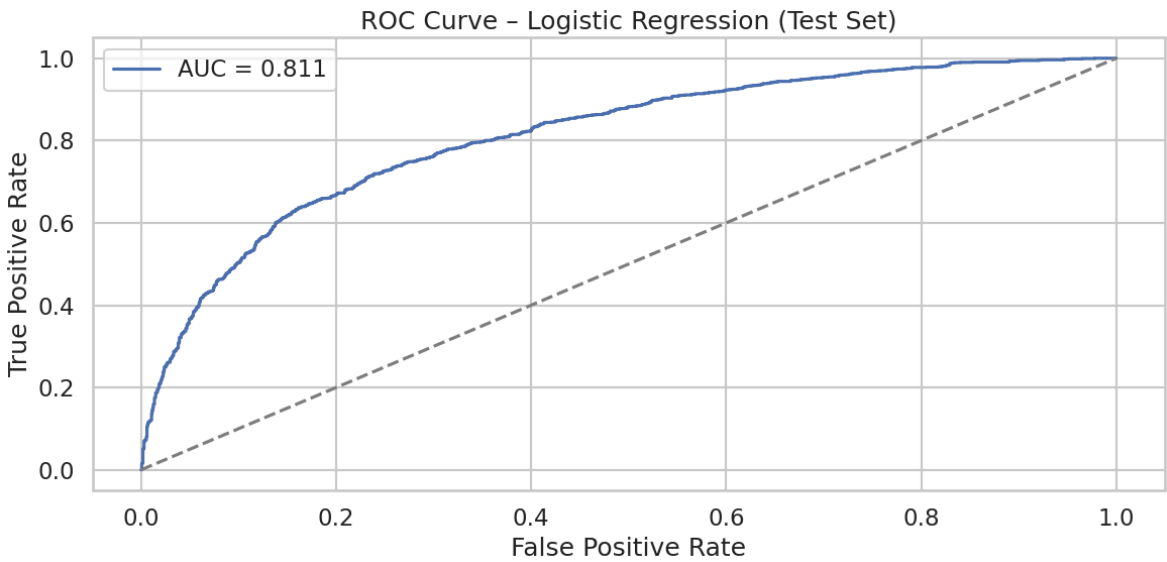


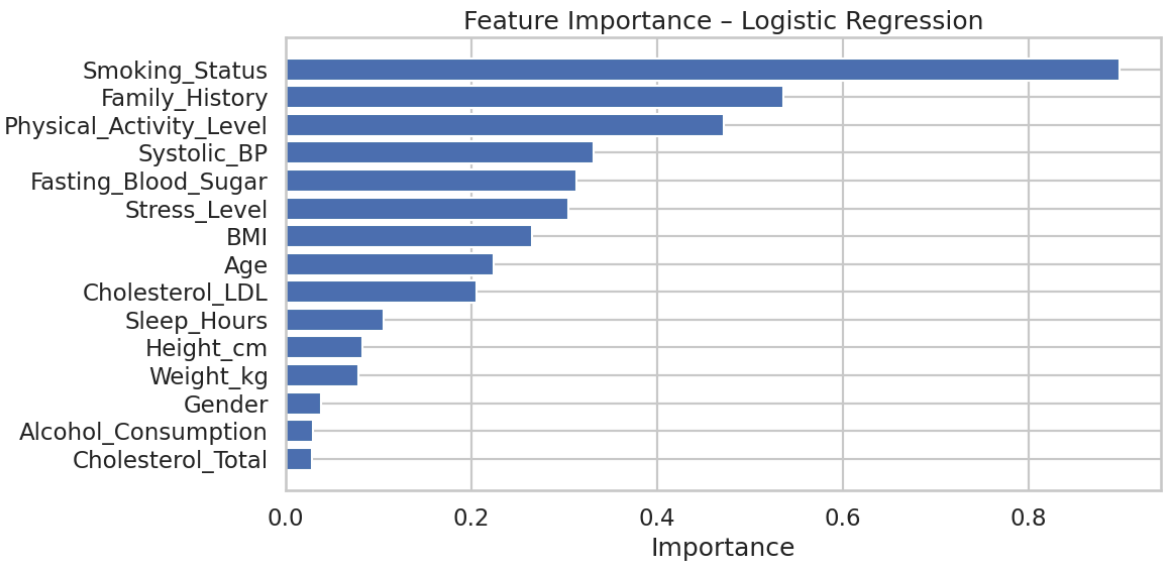
Logistic Regression

In [15]:
```
pipeline_rf = Pipeline([
    ('scaler', StandardScaler()),
    ('model', LogisticRegression())
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_results = evaluate_cv(pipeline_rf, X_train, y_train, cv)

print("Logistic Regression (CV mean):")
for k, v in cv_results.items():
    print(f"{k.capitalize():15}: {v:.3f}")

y_pred_test, y_proba_test = train_and_predict(
    pipeline_rf, X_train, y_train, X_test
)

classes = np.unique(y_train)

plot_confusion_matrix(
    y_test, y_pred_test, classes,
    "Confusion Matrix — Logistic Regression (Test Set)"
)

plot_roc(
    y_test, y_proba_test, classes,
    "ROC Curve — Logistic Regression (Test Set)"
)

plot_feature_importance(pipeline_rf, feature_names, top_n=15, title="Feat

rf_result = ModelResult(
    name="Logistic Regression",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

results_list.append(rf_result)
```

```
Logistic Regression (CV mean):
Accuracy       : 0.736
Macro_f1       : 0.727
Macro_recall   : 0.725
<Figure size 800x600 with 0 Axes>
```

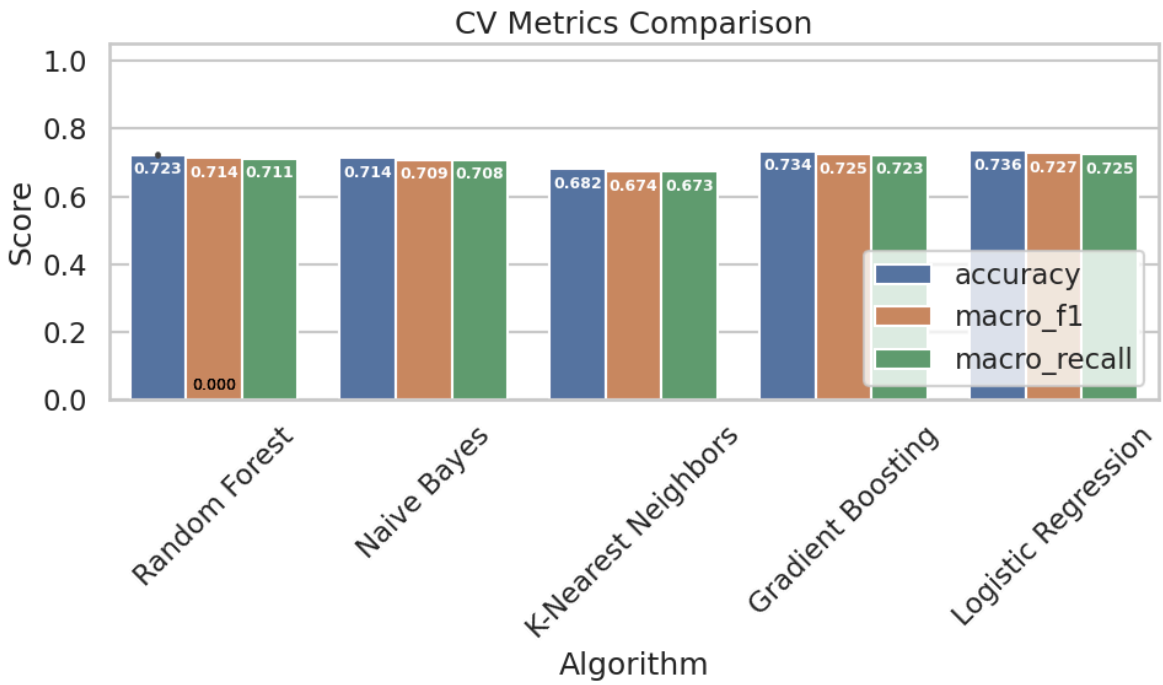






Model comparison

In [16]:
```
plot_cv_metrics(results_list)
```



Hyperparameters tuning

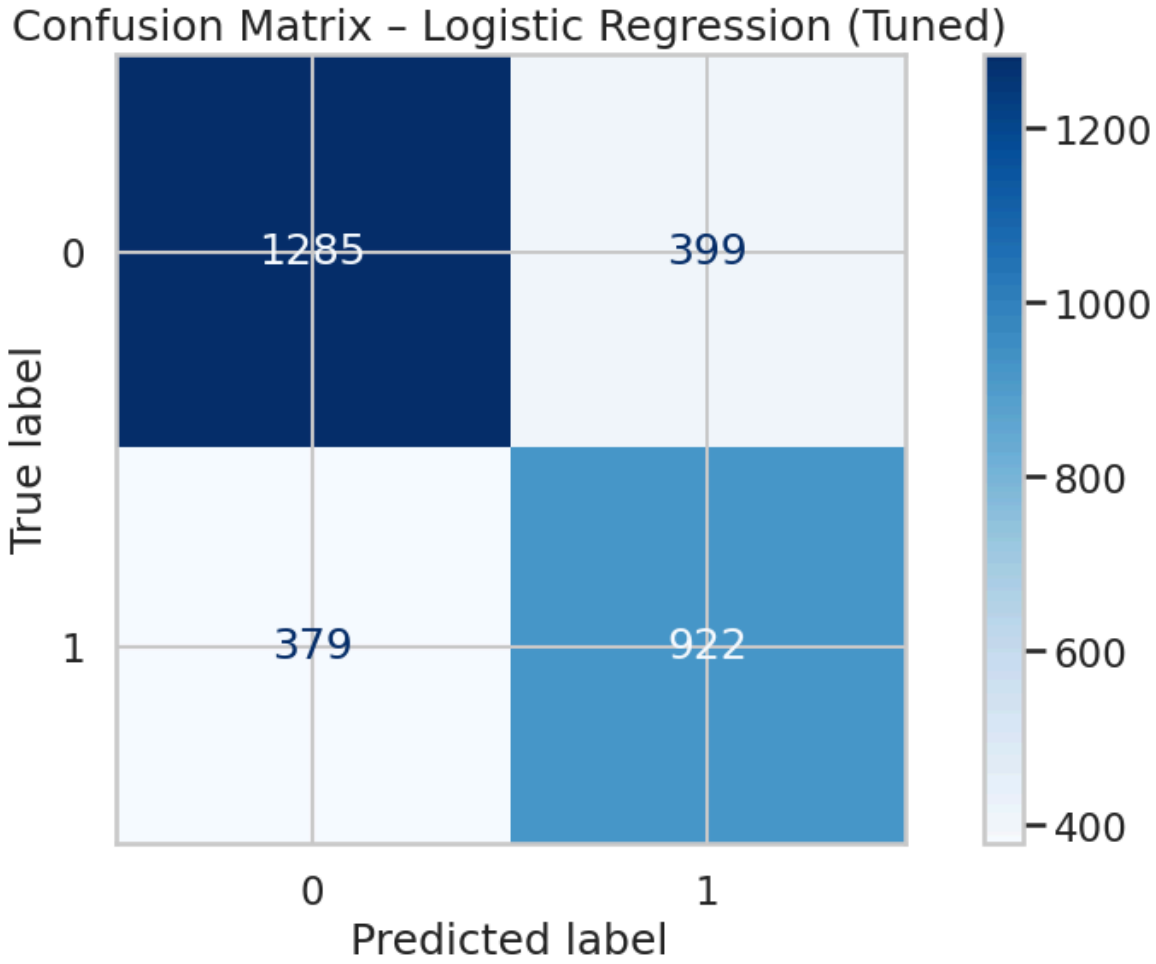
Zbog dobrih performansi, biramo *Random Forest* i *Gradient Boosting*.

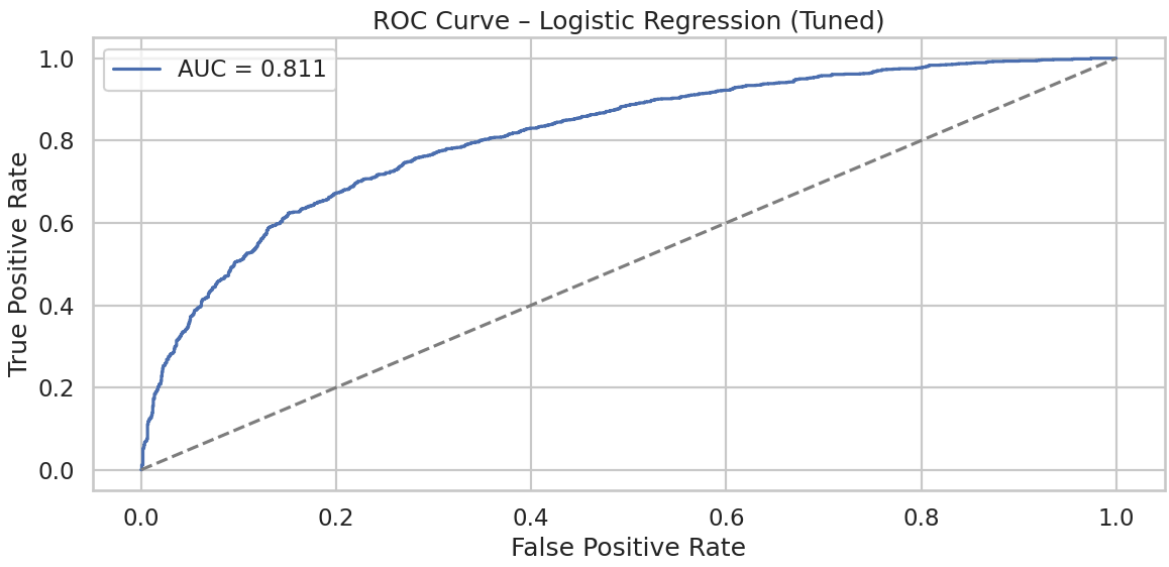
Logistic regresion Hyperparameters tuning

In [18]:
```
best_lr_model = grid_search_lr.best_estimator_

# Predict on test set
y_pred_test, y_proba_test = train_and_predict(best_lr_model, X_train, y_t
classes = np.unique(y_train)

# Plot confusion matrix and ROC
plot_confusion_matrix(y_test, y_pred_test, classes, "Confusion Matrix – L
```

```
plot_roc(y_test, y_proba_test, classes, "ROC Curve – Logistic Regression

# Store results
lr_result = ModelResult(
    name="Logistic Regression (Tuned)",
    cv_metrics={
        'accuracy': grid_search_lr.cv_results_['mean_test_score'][grid_se
        'macro_f1': grid_search_lr.cv_results_['mean_test_score'][grid_se
        'macro_recall': grid_search_lr.cv_results_['mean_test_score'][gri
    },
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

tuned_parameters_results_list.append(lr_result)
```

```
<Figure size 800x600 with 0 Axes>
```





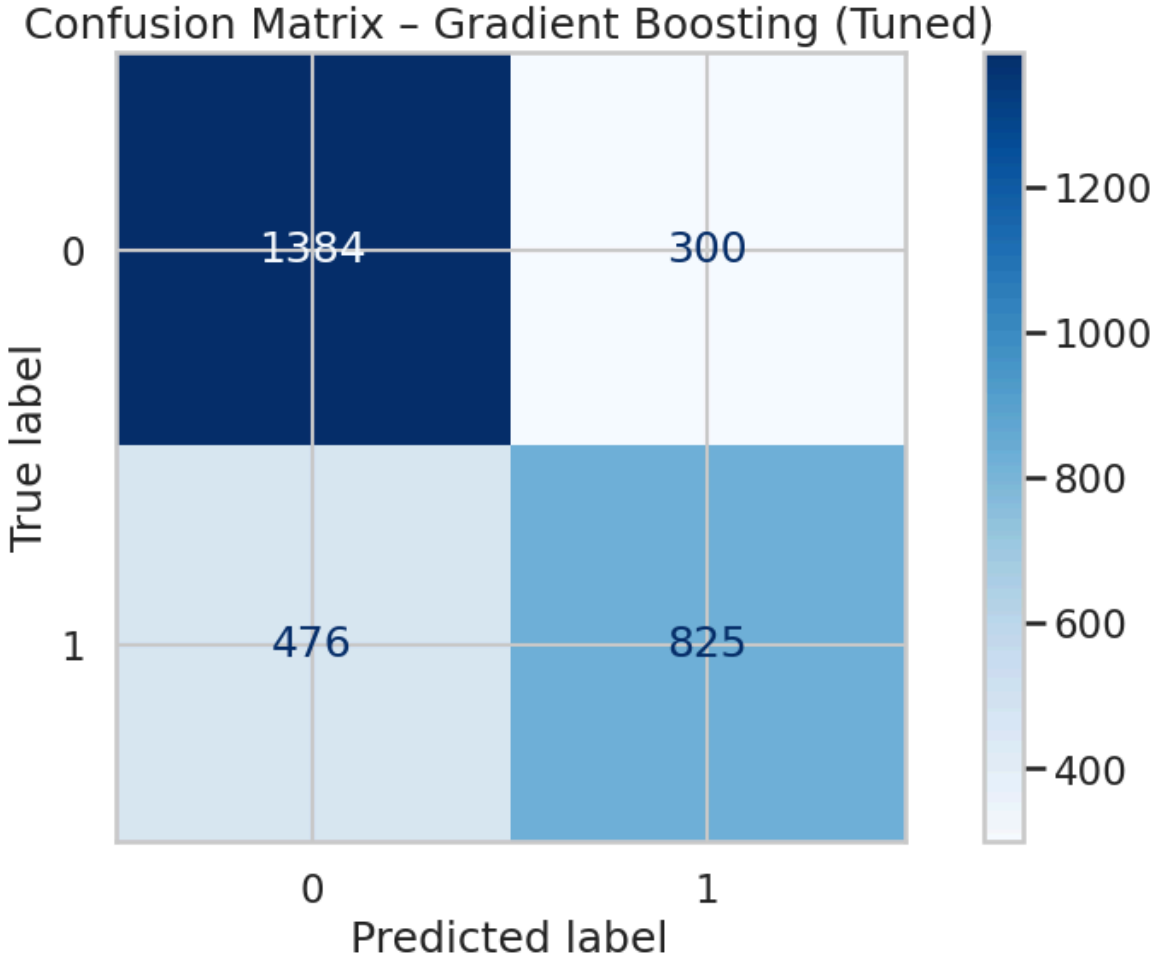
Gradient Boosting Hiperparameters tuning

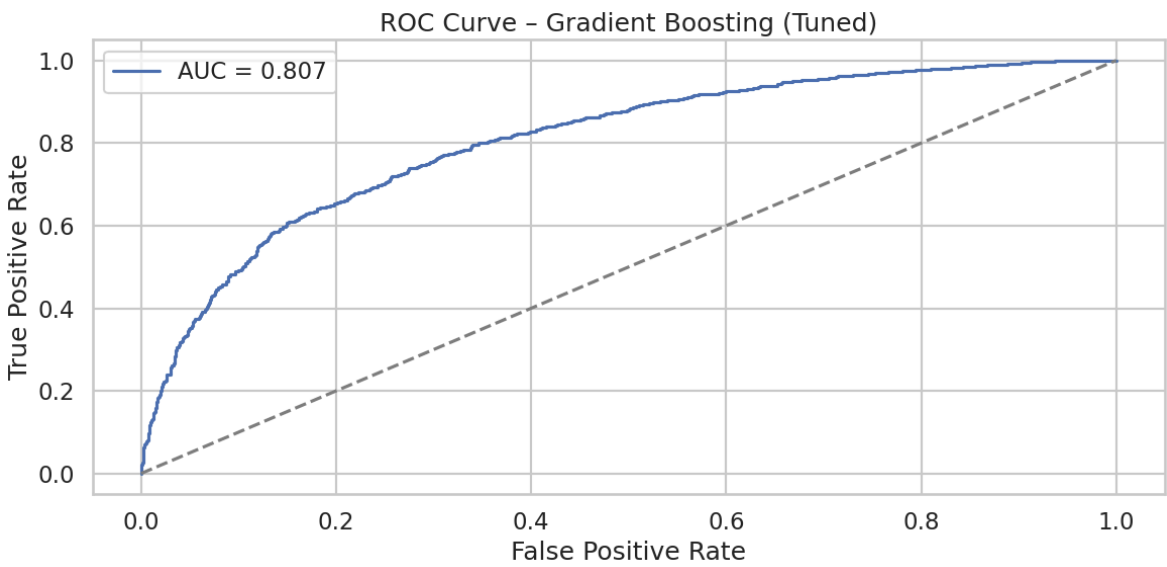
In [19]:
```
pipeline_gb = Pipeline([
    ('scaler', StandardScaler()),
    ('model', GradientBoostingClassifier(random_state=42))
])

param_grid = {
    'model__n_estimators': [100, 200, 300],
    'model__learning_rate': [0.01, 0.05, 0.1],
    'model__max_depth': [3, 5, 7],
    'model__subsample': [0.7, 1.0]
}

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

scoring = make_scorer(recall_score, average='macro')

grid_search = GridSearchCV(
    estimator=pipeline_gb,
    param_grid=param_grid,
    scoring=scoring,
    cv=cv,
    refit=True,
    n_jobs=-1,
    verbose=2
)

grid_search.fit(X_train, y_train)

best_gb_model = grid_search.best_estimator_

print("Best hyperparameters:", grid_search.best_params_)
print("Best CV macro F1 score:", grid_search.best_score_)
```

In [20]:
```
y_pred_test, y_proba_test = train_and_predict(best_gb_model, X_train, y_t
classes = np.unique(y_train)

plot_confusion_matrix(y_test, y_pred_test, classes, "Confusion Matrix – G
plot_roc(y_test, y_proba_test, classes, "ROC Curve – Gradient Boosting (T
```

```
cv_results = evaluate_cv(best_gb_model, X_train, y_train, cv)

gb_result = ModelResult(
    name="Gradient Boosting (Tuned)",
    cv_metrics=cv_results,
    y_pred_test=y_pred_test,
    y_proba_test=y_proba_test
)

tuned_parameters_results_list.append(gb_result)
```

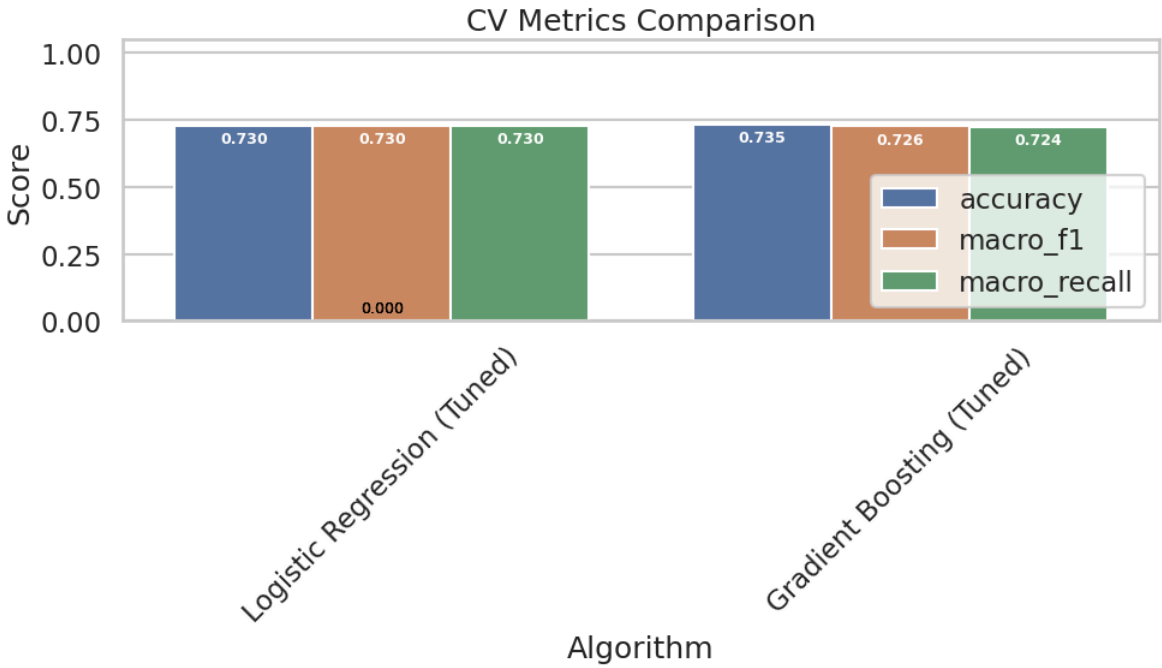
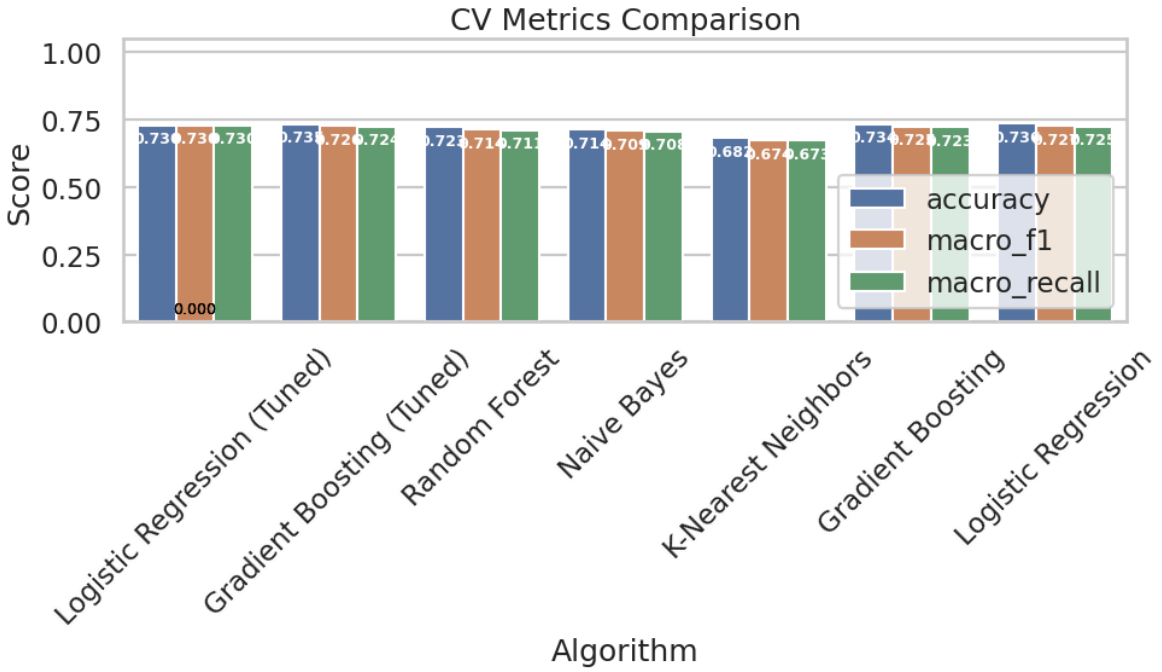
```
<Figure size 800x600 with 0 Axes>
```





Model comparison after hyperparameters tuning

In [21]:
```
plot_cv_metrics(tuned_parameters_results_list)
```



In [22]:
```
plot_cv_metrics(tuned_parameters_results_list + results_list)
```